



Prediction of protein interaction hot spots using rough set-based multiple criteria linear programming

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ABSTRACT

Protein–protein interactions are fundamentally important in many biological processes and it is in pressing need to understand the principles of protein–protein interactions. Mutagenesis studies have found that only a small fraction of surface residues, known as hot spots, are responsible for the physical binding in protein complexes. However, revealing hot spots by mutagenesis experiments are usually time consuming and expensive. In order to complement the experimental efforts, we propose a new computational approach in this paper to predict hot spots. Our method, Rough Set-based Multiple Criteria Linear Programming (RS-MCLP), integrates rough sets theory and multiple criteria linear programming to choose dominant features and computationally predict hot spots. Our approach is benchmarked by a dataset of 904 alanine-mutated residues and the results show that our RS-MCLP method performs better than other methods, e.g., MCLP, Decision Tree, Bayes Net, and the existing HotSpot database. In addition, we reveal several biological insights based on our analysis. We find that four features (the change of accessible surface area, percentage of the change of accessible surface area, size of a residue, and atomic contacts) are critical in predicting hot spots. Furthermore, we find that three residues (Tyr, Trp, and Phe) are abundant in hot spots through analyzing the distribution of amino acids.

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1. Introduction

Protein–protein interactions are fundamental for most biological processes such as signal transduction, regulation, and hormone–receptor interactions. In addition, disrupting or modulating the formation of protein complexes with small molecules will afford golden opportunities for therapeutic applications such as new medicine development and vaccine design. Therefore in the past years, we have seen many studies on the biochemical characteristics of protein interfaces, e.g., electronic interaction, shape complementary, hydrogen bonds, and salt bridges (Chakrabarti and Janin, 2002; Ofra and Rost, 2003a, 2003b; Jones and Thornton, 1997; McCoy et al., 1997; Norel et al., 1999; Schueler and Margalit, 1995; Sheinerman et al., 2000; Xu et al., 1997).

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Alanine-scanning mutagenesis is an effective approach for determining what residue accounts for the binding free energy of complexes (Wells, 1991) by measuring the change in the free energy of binding ($\Delta\Delta G$). Systematic studies have shown that only a small fraction of amino acid residues make a dominant contribution to the binding free energy of complexes. These residues have been termed as “hot spots” and they will cause substantial drop in the binding free energy when mutated to alanine (Bogan and Thorn, 1998; Clackson and Wells, 1995; DeLano, 2002). Other studies have reported that hot spots are usually surrounded by energetically less important residues, which form a hydrophobic O-ring to shelter hot spots from bulk solvent (Bogan and Thorn, 1998; Moreira et al., 2007a, 2007b). In addition, it has been found that hot spots are often located in the densely packed central clusters (Keskin et al., 2005).

Experiments on mutagenesis are usually costly and time consuming. Instead, computational method can predict hot spots in an effective way and may provide targets for further biological experiments. Many groups have proposed computational approaches for predicting hot spots based on biophysical and biochemical properties (Darnell et al., 2007; Cho et al., 2009) or free

energy calculation (Kortemme and Baker, 2002; Lafont et al., 2007; Almlof et al., 2006; Verkhrivker, 2004; Moreira et al., 2007a, 2007b). It has also been reported that hot spots can be predicted directly from sequence (Ofra and Rost, 2007, 2003a, 2003b). Despite these developments in prediction methods, simple and accurate approaches are still in pressing need.

Here, we apply a hybrid model, which combines rough sets (Pawlak, 1982) and the multiple criteria linear programming (MCLP) method (Kou et al., 2003; Shi, 2001), to predict hot spots. The rough set-based multiple criteria linear programming (RS-MCLP) method is initially proposed by Zhang et al. (2008) and has been used to analyze the medical diagnosis and prognosis with high performance (Zhang et al., 2009). As the first step, we collect 904 residues with $\Delta\Delta G$ value assayed experimentally. Secondly, nine important features are selected from fourteen features based on the rough sets theory. Thirdly, the MCLP method is applied to predict hot spots and we demonstrate that the RS-MCLP method outperforms the MCLP method and some well-known classification tools, e.g., Decision Tree (Quinlan, 1986) and Bayes Net (Jensen, 1996). In addition, we show that our RS-MCLP method outperforms an existing HotSpot database (Guney et al., 2008). Finally, we reveal several biological insights through our computational prediction and further analysis.

2. Materials and methods

2.1. Data set

We collect 28 protein complex structures from the Protein Data Bank (Bernstein et al., 1978; Berman et al., 2000) and these protein complexes are listed in Table 1. To ensure the high quality of gold standard dataset, we require that these X-ray crystal structures have higher resolution ($\leq 3 \text{ \AA}$) and sequence identity is less than 35%. We limit the number of mutations at equivalent sequence positions, which are deduced from BLAST (Basic Local Alignment Search Tool) (Altschul et al., 1990, 1997). A dataset of 904 alanine-mutated residues derived from 28 protein–protein complexes is obtained from the Alanine Scanning Energetics Database (ASEdb) (Thorn and Bogan, 2001), the Binding Interface Database (BID) (Fischer et al., 2003), and previous publications (Lang et al., 2000; Fasolini et al., 2003; Piehler and Schreiber, 1999; Van Petegem et al., 2008; Desrosiers and Peng, 2005; Eletr and Kuhlman, 2007; Gardsvoll et al., 2006; Gelinas et al., 2004). There are 107 hot spots ($\Delta\Delta G \geq 2.0 \text{ kcal mol}^{-1}$ when mutated to alanine) and 797 non-hot spots ($\Delta\Delta G < 2.0 \text{ kcal mol}^{-1}$). A detailed list of each mutation with its respective $\Delta\Delta G$ is reported as Supplementary Material. We note that our dataset is larger than those used in previous work (Darnell et al., 2007; Cho et al., 2009).

2.2. Residue features

Based on the previous studies about hot spots, a wide variety of features are collected, including the change of accessible surface area (ASA), percentage of the change of ASA, percentage of ASA for polar atoms in a residue, percentage of ASA for non-polar atoms in a residue, size of a residue, chemical nature (polar, non-polar, and charged), secondary structure (helix, sheet, and random coil), hydrogen bonds, salt bridges, and atomic contacts. In total, the 14 features are collected and listed in Supplementary Material. Subsequently, we do feature selection by using the rough sets theory (see Section 2.3.1). As a result we find that some features are insignificant to predict hot spots. Finally, we select 9 features that can generate robust predictive models. In the following we describe and explain the fourteen features.

Table 1

The 28 protein complexes used in our dataset.

PDB	Molecule 1	Molecule 2
1A4Y	Angiogenin	RNase inhibitor
1AHW	Immunoglobulin Fab 5G9	Tissue factor
1AWC	GA binding protein (GABP) alpha	GA binding protein beta 1
1BRS	Barnase	Barstar
1BXI	Colicin E9 immunity protein	Colicin E9 Dnase
1CBW	Chymotrypsin	BPTI trypsin inhibitor
1C4Z	Ubiquitin-protein ligase E3A	Ubiquitin conjugating enzyme E2
1DFJ	RNase inhibitor	RNase A
1DKG	Nucleotide exchange factor Grpe	Molecular chaperone Dnak
1DNA	Coagulation factor VIIa	Tissue factor
1DVF	Idiotopic antibody FV D1.3	Anti-idiotopic antibody FV E5.2
1DX5	Thrombin	Thrombomodulin
1F47	Cell division protein FtsZ	Cell division protein ZipA
1FC2	Fragment B of protein A	Immunoglobulin Fc
1GC1	Envelope protein gp120	CD4
1IO4	Runt-related transcription factor 1	Core-binding factor (CBF) beta
1JDH	Beta-catenin	Tcf4
1JRH	Antibody A6	Interferon-gamma receptor
1JTG	TEM-1 beta lactamase	Beta lactamase inhibitor protein
1KTZ	Transforming growth factor beta 3	TGF-beta Type II Receptor
1VFB	IgG1-kappa D1.3 Fv	Hen egg lysozyme
1TOJ	Voltage-gated calcium channel	Voltage-dependent L-type calcium
	Subunit Beta2aChannel	Alpha-1C subunit
2HYM	Soluble IFN alpha/beta recepto	Interferon alpha-2
3BPL	IL-4	IL-4R gamma
3BUK	Neurotrophin-	Gp75
3BT2	Urokinase-type plasminogen activator	Urokinase plasminogen activator surface receptor
3HFM	Ig FAB fragment HyHEL-10	Hen egg lysozyme
3HHR	Human growth hormone	Human growth hormone receptor

2.2.1. Solvent accessible surface area

The ASA of each residue is calculated using the program NACCESS (Hubbard and NACCESS, 1993) with a probe ball radius of $1.4 \text{ \AA}$. The ΔASA is the ASA change of a residue of the protein from monomer state to complex formation. As for the calculation of ASA, all water molecules in PDB files are ignored in our analysis:

$$\Delta\text{ASA} = \text{ASA}_{i\text{Und}} - \text{ASA}_{i\text{Bnd}}$$

where $\text{ASA}_{i\text{Und}}$ is the solvent accessible surface area for a residue i before subunits binding; $\text{ASA}_{i\text{Bnd}}$ is the solvent accessible surface area for the same residue in the complex. The $\Delta\text{ASA}\%$ for a residue is calculated as follows:

$$\Delta\text{ASA}\% = (\text{ASA}_{i\text{Und}} - \text{ASA}_{i\text{Bnd}}) / \text{ASA}_{i\text{Und}} \times 100\%$$

The $\Delta\text{ASA}\%$ has been described in previous study (Cho et al., 2009). Additionally, the ASA of polar or non-polar atoms for a residue in bound state is calculated using the NACCESS program. The percent of each ASA has also been analyzed.

2.2.2. Chemical nature and relative size of a residue

The residues are divided into three types (polar, non-polar, and charged) according to their chemical natures. Polar residues contain Asn, Cys, Gln, Gly, Ser, Thr, and Tyr. Non-polar residues are composed of Ala, Ile, Leu, Met, Phe, Pro, Trp, and Val. Charged residues include Arg, Asp, Glu, His, and Lys. The number of side-chain heavy atoms (C, S, N, and O) denotes the relative size of a residue.

2.2.3. Secondary structure

Here, we focus on three types of secondary structures: helix, sheet, and random coil. The regions of the protein's backbone that are not part of helices or sheets are defined as irregular secondary structures, for example, random coil.

2.2.4. Contact of atoms

Usually the protein interactions are chemically mediated by interacting atom pairs in which an atom from one protein may come in contact with other atom(s) from the partner protein. The interaction between atoms is defined as follows. The atoms are flagged as interacting atoms when the distance between any two atoms, one from each side, is less than the sum of their corresponding van der Waals radii plus 0.5 Å (Tsai et al., 1996).

2.2.5. Hydrogen bonds and salt bridges

We also investigate the hydrogen bonds across protein–protein interfaces. If the distance between donor atoms and acceptor atoms from different subunits is less than 3.5 Å, two atoms are considered to form a hydrogen bond. Moreover, we analyze salt bridges across interfaces. According to the distance between a positively charged atom and a negatively charged atom from each side of the interface, the salt bridge is evaluated. If the distance is under 4.5 Å, the pair is flagged as a salt bridge (Keskin et al., 2005).

2.3. Theoretical method

2.3.1. Rough sets-based feature selection approach

The rough sets theory is a powerful mathematical analysis method developed by Pawlak (1982) and is typically used to deal with fuzzy and uncertain information. In the past 20 years, applications of the rough sets theory have increased rapidly (Bhatt and Gopal, 2005; Kryszkiewicz, 1998). Moreover, attribute reduction is of importance in rough sets, which can efficiently reduce the redundancy in the attribute set (Bhatt and Gopal, 2005; Swiniarski and Skowron, 2003).

Suppose that two nonempty finite sets U and A are in a decision system or decision table, which denotes $A=(U,A)$, where U is the universe of objects and A is a set of attributes. For each attribute $a \in A$, we define V_a as the set of values of a , called the domain of attribute a .

A binary relation $R \in X \times X$, which may be reflexive (an object is in relation with itself xRx), symmetric (if $xRy \in yRx$), or transitive (if xRy and $yRz \rightarrow xRz$) denotes an equivalence relation. The equivalence class of an element $x \in X$ comprises all objects $y \in X$ such as xRy .

In an information system $A=(U,A)$, for any subset B belonging to A , an indiscernibility relation $IND(B)$ is defined as follows: $IND(B)=\{(x,y) \in U^2 | \forall a \in B, a(x)=a(y)\}$. If $(x,y) \in IND(B)$, thus two objects, x and y are indiscernible by the set of attributes B in A . Then, the equivalence class of x in relation $IND(B)$ is denoted as $[x]_{IND(B)}$. The equivalence class of $IND(B)$ is called elementary set in B . The elementary set is the partitions of universe U . The subsets that are most interesting have the same value of the outcome attribute (Komorowski et al., 1999).

In addition, the attribute $a(a \in A)$ can be deleted for $\forall a \in A$ if the equation $IND(A)=IND(A-\{a\})$ holds. In other words, the attribute a is reducible; otherwise, the attribute a is irreducible. Similarly, the attribute set A should be independent; otherwise, it is correlative and reducible. Further, if an attribute subset $B(B \subseteq A)$ is independent and $IND(B)=IND(A)$, set B is a reduction of set A . In addition, set B is the minimal reduction if it includes the fewest number of elements in all reduction attribute subsets.

To sum up, a rough set is an effective tool for treating data and has many advantages. First, prior knowledge is not needed for a rough set; and only the information in the data set is enough for

knowledge discovery. Second, on the basis of equivalent relation and indiscernibility relation, a rough set can effectively reduce redundant information and gain the minimal reduction of knowledge or attributes. Moreover, it can express uncertain information effectively. Third, a rough set can avoid the effects of missing values in a data set. Finally, a rough set can rapidly and efficiently process a lot of data with many attributes or variables.

2.3.2. MCLP method for classification

A general two-group classification problem solved by multiple criteria linear programming was developed by Shi (2001). The basic concept can be stated as follows.

Given a set of n variables or attributes in database $A=(a_1, a_2, \dots, a_n)$, let $A_i=(a_{i1}, a_{i2}, \dots, a_{im}) \in R^n$ be the development sample of data for the variables, where $i=1, 2, \dots, l$, and l is the total number of samples in the dataset. We need to choose a boundary value b to separate the two groups: G_1 (positives) and G_2 (negatives), that is

$$\begin{aligned} A_i X &\leq b, & A_i &\in G_1; \\ A_i X &\geq b, & A_i &\in G_2; \end{aligned} \quad (1)$$

here, A_i is the vector value of the subset of the variables, and each alternative or attribute data belongs to a certain group: G_1 or G_2 .

To measure the separation of G_1 and G_2 , two measurements are considered. Freed and Glover (1981, 1986) proposed that α_i be the overlapping degree for case A_i , and β_i be the distance of case A_i from its adjusted boundary. We want to minimize the sum of α_i and maximize the sum of β_i simultaneously, and then the initial linear programming can be modified as

$$\begin{aligned} &\text{Minimize } \sum_i \alpha_i \text{ and maximize } \sum_i \beta_i \\ &\text{Subject to} \end{aligned}$$

$$\begin{aligned} A_i X &= b + \alpha_i - \beta_i, A_i \in G_1; \\ A_i X &= b - \alpha_i + \beta_i, A_i \in G_2; \end{aligned} \quad (2)$$

where A_i is given, X and b are unrestricted, and $\alpha_i, \beta_i \geq 0$.

To improve computational performance, the compromise solution approach (Shi, 2001; Shi and Yu, 1989) has been implied to reform the above model. For an optimal solution, the best trade-off between minimizing the overlapping degree and maximizing the distance departed from boundary is systematically considered.

We assume that the ideal value of $-\sum_i \alpha_i$ is α^* ($\alpha^* > 0$), and the ideal value of $\sum_i \beta_i$ is β^* ($\beta^* > 0$). Then, if $-\sum_i \alpha_i > \alpha^*$, we define the regret measure as $-d_\alpha^+ = \alpha^* + \sum_i \alpha_i$ ($d_\alpha^+ \geq 0$); otherwise, it is 0. If $-\sum_i \alpha_i < \alpha^*$, the regret measure is also written as $d_\alpha^- = \alpha^* - \sum_i \alpha_i$ ($d_\alpha^- \geq 0$); otherwise, it is 0. Then, we have Eq. (1) $\sum_i \alpha_i + \alpha^* = d_\alpha^- + d_\alpha^+$, (2) $\sum_i \alpha_i + \alpha_i = d_\alpha^- + d_\alpha^+$. Similarly, we derive $\beta^* - \sum_i \beta_i = d_\beta^- - d_\beta^+$, $|\beta^* - \sum_i \beta_i| = d_\beta^- + d_\beta^+$, $d_\beta^- \geq 0$, and $d_\beta^+ \geq 0$. Thus, the evolved MCLP for a two-group separation model is the following:

$$\begin{aligned} &\text{Minimize } d_\alpha^- + d_\alpha^+ + d_\beta^- + d_\beta^+ \\ &\text{Subject to} \\ &\alpha^* + \sum_i \alpha_i = d_\alpha^- + d_\alpha^+; \\ &\beta^* - \sum_i \beta_i = d_\beta^- - d_\beta^+; \\ &A_i X = b + \alpha_i - \beta_i, A_i \in G_1; \\ &A_i X = b - \alpha_i + \beta_i, A_i \in G_2; \end{aligned} \quad (3)$$

where A_i , α^* and β^* are given, X and b are unrestricted, and $\alpha_i, \beta_i, d_\alpha^-, d_\alpha^+, d_\beta^-, d_\beta^+ \geq 0$.

Due to its simplicity, flexibility, and good generalization, the MCLP model is one of the most popular classification models (Kou et al., 2003; Zhang et al., 2008, 2009; Kou et al., 2005; Zheng et al., 2004).

2.3.3. A rough set-based MCLP method

In this paper, we apply the rough set-based multiple criteria linear programming method to predict hot spots. In biological terms, there are many attributes for measuring the characteristics of proteins. Thus it is necessary and beneficial to use rough sets for feature selection first, and then model with the reduced feature sets to achieve high predictive performance. The other reason is that the integration of the rough sets theory and the MCLP method is expected to work well. On the one hand, a rough set lacks fault tolerance and has poor generalization when used in a new data case. Nevertheless, the MCLP model overcomes the shortcomings of rough sets. On the other hand, the MCLP model can only deal with continuous and discrete data of a numeric type, and cannot reduce the dimensions of the input information space. However, rough sets can remove redundant information and gain a better dimensionality reduction. According to their characteristics of the complementarities, the combination of rough sets and MCLP achieves a new hybrid model. In predicting hot spots, the rough set serves as a prefix system which is responsible for data preprocessing, and MCLP is then applied to predict hot spots.

The algorithm of rough set-based MCLP model for classification is as follows:

Input: Input data and set parameters

$A = (a_1, a_2, \dots, a_n)$, Sample observation of data for the variable

$Y = \{0, 1\} \in R^m$, Target variable

b , Initial boundary value

α^* , Negative sum of the overlapping degree

β^* , Sum of the distance where A_i departed from its adjusted boundary b

Output: Output the results of the rough set-based MCLP model

$X = (x_1, x_2, \dots, x_m) \in R^m (m \leq n)$, Weights of the minimal attribute set

$\alpha_i (1 \leq i \leq l)$, The overlapping degree of each case

$\beta_i (1 \leq i \leq l)$, The distance departed from the boundary of each case

Processing flow: Rough set-based MCLP model for classification

Step 1: Preprocess data and compute the minimal attribute set using rough sets theory.

Step 2: Divide the data set into training set and test set, train and learn the MCLP model on training set, and test the model on the test set.

Step 3: Evaluate the predictive performance of models.

2.4. Evaluation of predictive performance

The predictive performance is evaluated by the following measures:

$$\text{Precision (P)} = \text{TP} / (\text{TP} + \text{FP})$$

$$\text{Recall (R)} = \text{TP} / (\text{TP} + \text{FN})$$

$$F1 = 2PR / (P + R)$$

$$\text{MCC} = (\text{TP} \times \text{TN} - \text{FP} \times \text{FN}) / \sqrt{(\text{TP} + \text{FN})(\text{TP} + \text{FP})(\text{TN} + \text{FP})(\text{TN} + \text{FN})}$$

where TP, FP, TN, and FN refer to the number of true positive, false positive, true negative, and false negative, respectively. Precision (P) is the fraction of predicted hot spots that are true hot spots, recall (R) is the fraction of true hot spots that are correctly predicted, and the F1 score is a measure to balance precision and recall rates.

In our study, there are 107 hot spots and 797 non-hot spots thus the dataset is imbalanced. Therefore another evaluation measure, the Matthew's correlation coefficient (MCC), is used to avoid the negative impacts of imbalanced dataset. The value of MCC is between -1 and 1 , and higher MCC corresponds to better

predictive performance. The MCC equals -1 if the predictions are completely contrary to the real value, 1 if the predictions are completely correct, and 0 if the predictions are randomly produced.

3. Results

3.1. Comparison with alternative methods

When the hot spots are defined as the residues with $\Delta\Delta G \geq 2.0$ kcal mol $^{-1}$, there are 107 hot spots and 797 non-hot spots. As the first step we select 9 features that can improve predictive ability from 14 features (see Supplementary Material) using rough sets. These 9 selected features are ΔASA , $\Delta\text{ASA}\%$, percentage of ASA for polar atoms in a residue, percentage of ASA for non-polar atoms in a residue, size of a residue, non-polar (chemical nature), helix (secondary structure), sheet (secondary structure), and atomic contacts, respectively. They contain important information and the combination of nine features can distinguish hot spots from non-hot spots. Then, the MCLP model is used to predict protein interaction hot spots. In our study, 50 hot spots and 50 non-hot spots are selected as training set and the remaining 804 residues are regarded as test set. The whole experiment is performed 10 times. Finally, we calculate the average value of recall, precision, F1 score, and MCC.

To validate the predictive performance of our method, we first compare our RS-MCLP method with the MCLP method. The results are reported in Table 2. The RS-MCLP method correctly predicts hot spots with $R=0.76$, $P=0.18$, $F1=0.28$, and $\text{MCC}=0.27$. This indicates that 76% of the true hot spots in the data set are correctly predicted, and 18% of predicted hot spots are correctly classified. Furthermore, MCLP only correctly predicts hot spots with $R=0.48$, $P=0.17$, $F1=0.25$, and $\text{MCC}=0.19$. In comparison, the F1 score of RS-MCLP is 3% points more than that of MCLP, and MCC for RS-MCLP relative to MCLP has increased with 8% points. These results show that RS-MCLP outperforms MCLP in predicting hot spots.

We then compare our RS-MCLP method with other traditional machine learning methods such as Decision Tree and Bayes Net. Comparison results are listed in Table 3. In summary, RS-MCLP method shows the best performance with $F1=0.28$. The F1 scores of Decision Tree and Bayes Net are 0.26 and 0.27, respectively. The F1 score of RS-MCLP is slightly higher than the two traditional methods. In addition, the scores for precision, recall, and the MCC of our method are comparable to traditional methods.

Finally, we compare the predicted hot spots by the RS-MCLP method with an existing hot spots database. HotSprint (Guney et al., 2008) is known as a database of computational hot spots in protein interfaces. HotSprint identifies hot spots based on conservation score, complex accessible solvent area (ASA), and ΔASA .

Table 2

Comparison of predictive performance for MCLP and RS-MCLP method.

Model	Recall	Precision	F1 score	MCC
MCLP	0.48 ± 0.1	0.17 ± 0.02	0.25 ± 0.02	0.19 ± 0.03
RS-MCLP	0.76 ± 0.02	0.18 ± 0.01	0.28 ± 0.02	0.27 ± 0.02

Table 3

Comparison of predictive performance for Decision Tree, Bayes Net, and RS-MCLP.

Model	Recall	Precision	F1 score	MCC
Decision Tree	0.72 ± 0.09	0.16 ± 0.03	0.26 ± 0.04	0.23 ± 0.05
Bayes Net	0.76 ± 0.05	0.16 ± 0.02	0.27 ± 0.03	0.25 ± 0.04
RS-MCLP	0.76 ± 0.02	0.18 ± 0.01	0.28 ± 0.02	0.27 ± 0.02

The conserved residues in interfaces, which have certain buried accessible solvent area (Δ ASA) and complex ASA thresholds, are marked as computational hot spots in HotSprint database (Guney et al., 2008). In the 904 residues dataset, HotSprint database only predicts 776 residues. To avoid the bias, we compare the predictive performance of RS-MCLP and HotSprint on the subset with 776 mutational residues. The comparison results are shown in Table 4. HotSprint correctly predicts hot spots with $R=0.31$, $P=0.32$, $F1=0.32$, and $MCC=0.23$; while RS-MCLP correctly predicts hot spots with $R=0.86$, $P=0.26$, $F1=0.40$, and $MCC=0.36$. The F1 score for RS-MCLP relative to HotSprint is 8 percentage points improved. Once more, an increase in MCC has 13% points. Our RS-MCLP has a higher recall value relative to HotSprint, which indicates that our RS-MCLP can predict hot spots in a more accurate way.

In summary, these results suggest that our RS-MCLP method has better predictive performance for hot spots. We note that all the

calculations and experiments are carried out on the same dataset, so the above comparisons are fair and meaningful.

3.2. Biological insights

We use rough sets to select the core of attributes, i.e., the dominant attributes that can capture the main properties of the experimentally derived hot spots. In our data set, these dominant attributes are listed as follows: Δ ASA, Δ ASA%, size of a residue, and atomic contacts. To validate the importance of these features, we train our RS-MCLP method excluding one feature at a time. We find that omission of each feature leads to some decrease in F1 and MCC score (see Table 5). In addition, we also perform statistical analysis to verify whether each feature mentioned above can distinguish hot spots from other residues. A nonparametric test Wilcoxon Two-Sample Test (rank sum test) has been carried out to examine whether two independent samples have significant difference. Results have shown that there is statistical significance between hot spots and other residues with P -value < 0.001 for four attributes, respectively (see Fig. 1).

We further analyze the distribution of amino acids in the whole data set and in hot spots. The number and percentage of each amino acid type are shown in Table 6. Enrichment of particular residue in hot spots shows the preference of that residue type in hot spots over the whole data set. We find that our results agree well with the previous studies (Bogan and Thorn, 1998; Lise et al., 2009), e.g., Tyr and Trp are among the most frequent residues in hot spots. However, we also find some interesting differences, e.g., hot spots are enriched in Phe in our data set (Arg in Bogan's study and Lys in Lise's study). Although Bogan's study is based on the whole ASEdb database, we have removed the redundant entries with sequence identify more than 35%. Lise et al. have collected the structure of the protein complex without considering the resolution while we

Table 4
Predictive results of HotSprint and RS-MCLP on 776 mutational residues.

Model	Recall	Precision	F1 score	MCC
HotSprint	0.31	0.32	0.32	0.23
RS-MCLP	0.86 $\pm$ 0.01	0.26 $\pm$ 0.01	0.40 $\pm$ 0.01	0.36

Table 5
Results for RS-MCLP on the subsets by deleting one feature every time.

Deleted feature	Recall	Precision	F1 score	MCC
Δ ASA	0.80 $\pm$ 0.05	0.16 $\pm$ 0.01	0.26 $\pm$ 0.02	0.25 $\pm$ 0.02
Δ ASA%	0.70 $\pm$ 0.08	0.16 $\pm$ 0.01	0.26 $\pm$ 0.02	0.23 $\pm$ 0.03
Size of a residue	0.82 $\pm$ 0.06	0.16 $\pm$ 0.01	0.26 $\pm$ 0.01	0.26 $\pm$ 0.02
Atomic contact	0.84 $\pm$ 0.04	0.15 $\pm$ 0.01	0.26 $\pm$ 0.01	0.25 $\pm$ 0.02

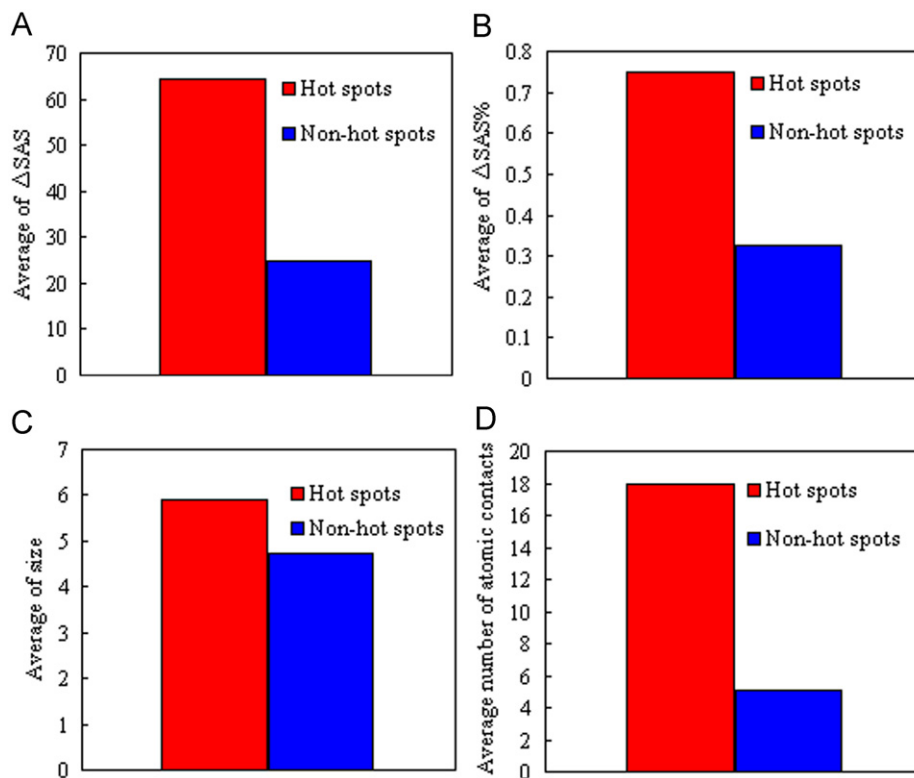


Fig. 1. Comparison between hot spots and non-hot spots regarding four dominant features: (A) the change of solvent accessible surface area, (B) the percentage of Δ ASA, (C) the size of a residue, and (D) atomic contacts.

Table 6
Distributions of amino acid types in our data set.

Mutated residue	In whole data set		Hot spots		Enrichment in hot spots
	(Number)	(%)	(Number)	(%)	
Arg	100	11.06	13	12.15	1.10
Asn	52	5.75	6	5.61	0.97
Asp	73	8.08	11	10.28	1.27
Cys	8	0.8	2	1.87	2.11
Gln	50	5.53	1	0.93	0.17
Glu	92	10.18	6	5.61	0.55
Gly	6	0.66	0	0	0
His	25	2.77	1	0.93	0.34
Ile	40	4.42	8	7.48	1.69
Leu	57	6.31	7	6.54	1.04
Lys	88	9.73	10	9.35	0.96
Met	18	1.99	1	0.93	0.47
Phe	25	2.77	7	6.54	2.37
Pro	17	1.88	0	0	0
Ser	67	7.41	1	0.93	0.13
Thr	58	6.42	3	2.80	0.44
Trp	31	3.43	9	8.41	2.45
Tyr	55	6.08	20	18.69	3.07
Val	42	4.65	1	0.93	0.20

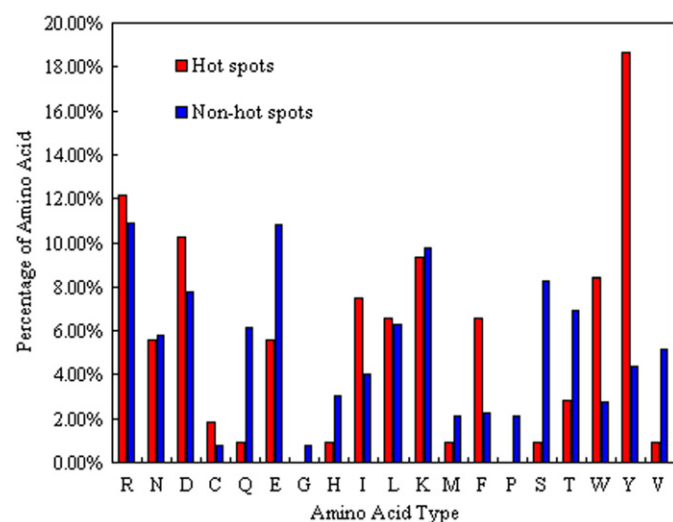


Fig. 2. Distributions of amino acid types in hot spots and non-hot spots.

retain the structure of the protein complex which is available (resolution ≤ 3 Å). We think that our statistics of enriched residues in hot spots are based on a more reliable dataset and the revealed differences with other methods need to be further explored.

We have also analyzed the distribution of each amino acid type in hot spots and non-hot spots (see Fig. 2). It is clear that 8 amino acid types are enriched in hot spots, 4 types of which (Ile, Leu, Phe, Tyr) are hydrophobic residues. Tyr and Trp are usually found in hot spots. Both of them are hydrophobic aromatic residues, and capable of forming one hydrogen bond. They can form aromatic π -interactions, and have large hydrophobic surface. Likewise, Phe also contributes aromatic π -interactions, and offers hydrophobic surface. Cys is enriched in hot spots from our data set, presumably owing to forming a disulfide bond with other Cys. It is worth noting that Asp is greatly enriched in hot spots than Glu. And it is also found that Asn, with a frequency of 5.61%, is more than six times as frequent in hot spots as Gln (0.93%). This differences may come from the side-chain conformational entropy (Bogan and Thorn, 1998).

4. Discussion

In this paper, we applied RS-MCLP approach for predicting hot spots. The basic idea of the MCLP model is to minimize the overlapping rate of the classes and maximize the distance between the classes (Kou et al., 2003; Shi, 2001). The MCLP model works well in dealing with classification problems. However, the MCLP model is not good at dimensional reduction and removing information redundancy, especially when there are many features. Rough sets can efficiently remove redundant information and choose a minimal attribute set. Additionally, rough sets can efficiently process data with numerous types of variables. In this study, we have shown that RS-MCLP model can predict hot spots with higher F1 and MCC scores.

The results of this method are compared with other classification methods such as Decision Tree and Bayes Net. Comparison results have shown that RS-MCLP method has better performance in predicting hot spots, with $R=0.76$, $P=0.18$, $F1=0.28$, and $MCC=0.27$. We have also compared the predictive performance of RS-MCLP approach with an existing hot spots database. Based on a subset including 776 mutational residues, again our RS-MCLP method has better predictive performance with $R=0.86$, $P=0.26$, $F1=0.40$, and $MCC=0.36$. In addition, we have revealed several biological insights based on our predictive results. We have found that four features play dominant roles in predicting hot spots. Furthermore we have found that Tyr, Trp, and Phe are abundant in hot spots through analyzing the distribution of amino acids.

We note that this is the first application of the rough set-based MCLP method to the prediction of proteins interaction hot spots. We will make an effort to improve the performance of this method in the future. For example, one can explore more features of surface residues such as sequence information, structural features, molecular interactions, and a hot spot residue's microenvironment. Moreover, kernel function can be introduced into the current RS-MCLP method.

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Appendix A. Supplementary materials

Supplementary data associated with this article can be found in the online version at doi:10.1016/j.jtbi.2010.10.021.

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